

Question Number	Answer	Additional guidance	Mark
3(a)(i)	An explanation that makes reference to the following: • magnesium ions to make chlorophyll so {photosynthesis can occur / light energy can be absorbed / glucose can be made} (1) • nitrates are needed to make {amino acids / proteins / polypeptides / DNA / RNA / nucleic acid} (1)	accept nucleotide / bases / chlorophyll / ATP	(2)

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	An answer that makes reference to one of the following: • {grow/ store} more plants in a given area / less space needed to {grow/ store} the same number of plants (1) • increases profit as increased yield for lower rent (1)	Accept space saving / increased yield in given area	(1)

Question Number	Answer	Additional guidance	Mark
3(b)(i)	The only correct answer is D 14.39% <i>A is not correct because that is the percentage difference for 1 spray with the higher concentration</i> <i>B is not correct because that is the percentage difference for 1 spray with the lower concentration</i> <i>C is not correct because that is the percentage difference for 2 sprays with the higher concentration</i>		(1)

Question Number	Answer	Additional guidance	Mark
3(b)(ii)	An answer that makes reference to one of the following: - increasing the number of treatments increases (fruit) firmness / positive correlation between number of treatments and (mean fruit) firmness (1) - increasing the concentration of calcium (ions / chloride) increases (fruit) firmness / positive correlation between calcium (chloride) concentration and (mean fruit) firmness (1)	accept fruit treated with 3800 mg dm⁻³ were firmer than the fruit treated with 950 mg dm⁻³ ignore references to 'more calcium chloride solution used'	(1)

Question Number	Answer	Additional guidance	Mark
3(b)(iii)	An explanation that makes reference to three of the following: • increasing the concentration of calcium (ions / chloride) increases (fruit) firmness (1) • (because calcium ions are) needed to make calcium pectate (1) • (forming) middle lamella (1) • more {calcium pectate / middle lamella} formed would mean the cells were (more) firmly held together (1)	Accept {calcium pectate / middle lamella} {increases strength of cell wall / holds cells together}	(3)